

Physical Profile Pullback and the Cross-Divisor Gram

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Abstract

The transition-native twin-prime route contains one residual profile for every squarefree divisor, but these profiles are generated by the same physical sequence. Treating them as an orthogonal direct sum therefore changes the operator before the first Cauchy estimate. This paper constructs the missing finite coupling map. A residue lift sends a common source on a finite support to every divisor residue space; the reciprocal emitter is then pulled back to that support. We prove an exact affine pullback identity, an exact gcd/lcm formula for cross-divisor residue lifts, and an exact frequency-intersection formula for the Hermitian Gram of the pulled-back emitters. In a complete period, off-diagonal terms survive precisely when the rational frequencies agree. The fixture with divisors 5, 7, 35, reciprocal primes 11, 13, 17, and $H = 40$ gives cross-Gram values 560 and 770 for the nested pairs, while the coprime pair has zero cross-Gram. These identities establish a real structural advance over a direct-sum emitter model, but they do not estimate the literal smooth V46 kernel. The prime shell, four-packet signs, asymptotic Gram bound, Gate B, and arithmetic L^2 credit remain open.

Claim level. PROVED_STRUCTURAL_L1 / CROSS_DIVISOR_COUPLING.

No arithmetic advance, fixed-atom credit, or strict 1/400 payment is claimed.

1 Introduction

The current analytic route to the twin-prime problem reduces a fixed-gap count to a transition scalar indexed by squarefree divisors. In the V46 formulation, the transition residual has the schematic form

$$\mathfrak{R}_x^{\text{AP}} = -H \sum_{Y_0 < d \leq U} \sum_{r \bmod d} A_d(r) \widehat{\mathcal{R}}_d(r), \quad (1)$$

where the reciprocal emitter is

$$A_d(r) = \frac{\mu(d) \log d}{d} \sum_{q \in \mathcal{Q}} \sum_{0 < |m| \leq dq/H} \psi\left(\frac{Hm}{dq}\right) \mathbf{1}_{r \equiv m\bar{q} \pmod{d}}, \quad (2)$$

and $\mathcal{R}_d$ is the arithmetic-progression profile after subtracting the divisor-dependent Euler correction [1].

TPC-212 isolated the divisor cut and the reciprocal occupancy map. In the natural direct sum of residue spaces, the emitter Gram is block diagonal, so unit-weight residuals can align all blocks. That is a valid interface obstruction, but it deliberately leaves open whether the literal residuals can be chosen independently [2, 3]. The next question is therefore more basic than an asymptotic estimate:

What is the exact operator obtained when all divisor profiles are generated from one common physical sequence before Cauchy and before the outer absolute value?

This paper answers that operator question. Let $\mathcal{U}$ be a finite set of physical indices. For each divisor d , the residue lift aggregates a common source by congruence classes. The additive transform and the emitter then pull the result back to $\mathcal{U}$. The total source term contains the sum of these pulled-back kernels, not an orthogonal sum. The cross-divisor Gram is governed by shared rational frequencies and by the gcd of the moduli.

The results are structural and finite. They make no claim about the shifted prime sequence, and the certificate uses unit reciprocal weights only. The main contributions are:

1. an exact common-source pullback identity that retains the divisor- dependent profile correction;
2. an exact Chinese-remainder formula for the cross blocks of the residue lift;
3. an exact frequency-intersection formula for the physical emitter Gram;
4. a finite certificate showing nonzero nested-divisor cross terms and explicitly separating this structural result from an asymptotic saving.

2 The common-source physical operator

2.1 Residue lifts and transforms

Let $\mathcal{D}$ be a finite family of positive moduli and let $\mathcal{U}$ be a finite set of integers. For a source vector $f = (f(u))_{u \in \mathcal{U}}$ define the residue lift

$$(C_d f)(a) = \sum_{\substack{u \in \mathcal{U} \\ u \equiv a \pmod{d}}} f(u), \quad a \in \mathbb{Z}/d\mathbb{Z}. \quad (3)$$

The unnormalized additive transform is

$$(\mathcal{F}_d g)(r) = \sum_{a \bmod d} g(a) e_d(ra), \quad e_d(x) = \exp(2\pi i x/d). \quad (4)$$

The notation deliberately matches the unnormalized transform in the V46 transition artifact.

For each $d \in \mathcal{D}$, let $A_d(r)$ be any finite complex emitter row and define its physical pullback by

$$K_d(u) = \sum_{r \bmod d} A_d(r) e_d(ru), \quad u \in \mathcal{U}. \quad (5)$$

The literal V46 emitter in (2) is one instance. No estimate is assumed in the following identities.

The divisor-dependent profile correction is most conveniently written as a source correction. Let $v(u)$ be the common physical sequence and let $b_d(u)$ be a correction that may depend on d through the local Euler profile. Set

$$R_d = C_d(v - b_d). \quad (6)$$

Thus $R_d(a)$ is the residue-class sum of the same v with a different divisor correction, exactly as in the AP residual in V46.

Theorem 2.1 (common-source pullback). *For finite $\mathcal{U}$, $\mathcal{D}$, source v , corrections b_d , and emitter rows A_d , one has the exact bilinear identity*

$$\sum_{d \in \mathcal{D}} \sum_{r \bmod d} A_d(r) (\mathcal{F}_d R_d)(r) = \sum_{u \in \mathcal{U}} v(u) K(u) - \sum_{d \in \mathcal{D}} \sum_{u \in \mathcal{U}} b_d(u) K_d(u), \quad (7)$$

where

$$K(u) = \sum_{d \in \mathcal{D}} K_d(u). \quad (8)$$

Proof. For a fixed d , substitute (6) and (4):

$$\sum_r A_d(r) (\mathcal{F}_d R_d)(r) = \sum_r A_d(r) \sum_a \sum_{\substack{u \in \mathcal{U} \\ u \equiv a \pmod{d}}} (v(u) - b_d(u)) e_d(ra).$$

Whenever $u \equiv a \pmod{d}$, one has $e_d(ra) = e_d(ru)$. All sums are finite, so interchanging their order gives

$$\sum_r A_d(r) (\mathcal{F}_d R_d)(r) = \sum_{u \in \mathcal{U}} (v(u) - b_d(u)) \sum_r A_d(r) e_d(ru) = \sum_{u \in \mathcal{U}} (v(u) - b_d(u)) K_d(u).$$

Summing over d and collecting the common $v(u)$ term proves (7). $\square$

Remark 2.2. The second term in (7) is essential. It is the finite analogue of the V46 $\Delta_{d,z}$ correction and cannot be silently absorbed into v because it changes with d . The first term, however, uses one common source and the joint kernel $K = \sum_d K_d$.

3 The residue-lift cross blocks

The common-source statement already shows why a direct sum is not the physical operator. The next proposition identifies the lift cross blocks exactly.

Proposition 3.1 (gcd/lcm residue aliasing). *Let $d, e \geq 2$, let L be a positive multiple of $\text{lcm}(d, e)$, and take $\mathcal{U} = \{0, 1, \dots, L-1\}$. Then*

$$(C_d C_e^*)(a, b) = \frac{L}{\text{lcm}(d, e)} \mathbf{1}_{a \equiv b \pmod{\text{gcd}(d, e)}}. \quad (9)$$

Proof. The (a, b) entry counts solutions in the complete interval to $u \equiv a \pmod{d}$ and $u \equiv b \pmod{e}$. The Chinese remainder theorem says that the system is solvable if and only if the two residues agree modulo $g = \text{gcd}(d, e)$. If it is solvable, its solutions form one class modulo $L_0 = \text{lcm}(d, e)$. There are exactly L/L_0 such solutions in the chosen interval, which gives (9). $\square$

Corollary 3.2 (the lift is generally not orthogonal). *If $\text{gcd}(d, e) > 1$, then $C_d C_e^*$ has nonzero entries on every compatible residue pair. Thus a common physical source produces cross-divisor residue coupling before an emitter is applied.*

The statement is elementary but important for the route. TPC-212's direct sum is a useful obstruction space; it is not the result of applying the literal residue lifts to one common sequence. In a finite family containing a maximal modulus, the joint lift can also have fewer independent output coordinates than the sum of the individual row counts. The certificate below gives a concrete instance with $5 + 7 + 35 = 47$ output rows but rank 35.

4 The emitter pullback Gram

The residue lift is only one half of the coupling. The emitter can create or remove cross terms depending on its frequency support.

Theorem 4.1 (finite-interval pullback Gram). *For any finite support $\mathcal{U}$ and any $d, e \in \mathcal{D}$,*

$$\sum_{u \in \mathcal{U}} K_d(u) \overline{K_e(u)} = \sum_{r \bmod d} \sum_{s \bmod e} A_d(r) \overline{A_e(s)} \sum_{u \in \mathcal{U}} e\left(u \left(\frac{r}{d} - \frac{s}{e}\right)\right). \quad (10)$$

Proof. Insert (5) for K_d and the conjugate expansion for K_e . The resulting sum is finite, so the order of summation may be interchanged. The inner sum is precisely the finite exponential sum displayed in (10). $\square$

Theorem 4.2 (complete-period frequency intersection). *Let $L = \text{lcm}(d, e)$ and $\mathcal{U} = \{0, \dots, L-1\}$. Then*

$$\sum_{u=0}^{L-1} K_d(u) \overline{K_e(u)} = L \sum_{\substack{r \bmod d, s \bmod e \\ r/d \equiv s/e \pmod{1}}} A_d(r) \overline{A_e(s)}. \quad (11)$$

In particular, the cross Gram is zero if the two emitter frequency supports have no common rational frequency.

Proof. For each pair (r, s) , the inner exponential sum in (10) is a geometric series with ratio $e(r/d - s/e)$. Since $L(r/d - s/e)$ is an integer, the ratio is an L th root of unity. The sum over a complete period is L if the ratio is one, which is equivalent to $r/d - s/e \in \mathbb{Z}$, and is zero otherwise. Substitution into (10) proves the formula. $\square$

Corollary 4.3 (the physical Gram is not block diagonal in general). *Suppose there are $d \neq e$ for which the right-hand side of (11) is nonzero. Then the physical pullback Gram has a nonzero (d, e) entry. Consequently, replacing the common-source family by an orthogonal direct sum changes the quadratic form.*

Proof. The right-hand side is exactly the (d, e) Gram entry by Theorem 4.2. A block-diagonal matrix has zero entries between distinct divisor blocks, so a nonzero entry cannot be represented by that direct sum without an additional projection or cancellation theorem. $\square$

Remark 4.4 (what the theorem does not say). The existence of an off-diagonal Gram entry is not itself a saving. It can have either sign after the $\mu(d) \log d$ coefficients and packet signs are inserted, and it can increase a norm for an aligned vector. The theorem only identifies the exact object that a physical Gram estimate must control.

5 Exact finite certificate

5.1 Fixture and checks

The certificate uses the squarefree divisor family

$$\mathcal{D} = \{5, 7, 35\}, \quad \mathcal{U} = \{0, 1, \dots, 34\},$$

with cutoff $z = 3$, reciprocal primes $\mathcal{Q} = \{11, 13, 17\}$, and $H = 40$. Every q is a unit modulo every divisor, and the finite corridor satisfies $2|m| < d$ on all nonempty rows. The profile correction is the literal finite product difference $\Delta_{d,z} = P_d - B_{d,z}$ from V46.

Table 1: Exact unit-weight reciprocal occupancies.

d	$\#\{(q, m)\}$	occupancy summary	$\ A_d\ _2^2$	$2 \max m < d$
5	8	(0, 2, 2, 2, 2)	16	yes
7	10	(0, 1, 3, 1, 1, 3, 1)	22	yes
35	68	full 35-coordinate row in certificate	156	yes

Table 2: Exact complete-period pullback Gram.

pair (d, e)	period	common-frequency weight	cross Gram	normalized square
(5, 7)	35	0	0	0
(5, 35)	35	16	560	4/39
(7, 35)	35	22	770	11/78

Table 1 records the exact occupancy rows. The first row has zero frequency occupancy 0 because the finite corridor excludes $|m| \geq d$.

The residue-lift checker compares every cross block with the CRT formula. The joint lift has 47 rows and 35 columns. Since the $d = 35$ block is the identity on the chosen period, its rank is 35; hence the codomain has $47 - 35 = 12$ linear dependencies. This is a finite rank statement, not a claim about the dimension of an asymptotic profile space.

For the emitter pullbacks, the common-frequency products are

The values $560 = 35 \cdot 16$ and $770 = 35 \cdot 22$ follow directly from Theorem 4.2. The nonzero entries arise because a frequency $r/5$ or $s/7$ is represented by the corresponding frequency $7r/35$ or $5s/35$. The coprime pair (5, 7) has no nonzero common frequency in this fixture.

The Euler profiles also pass the two V46 structural invariants

$$\Delta_{d,z}(0) = 0, \quad \sum_{a \bmod d} \Delta_{d,z}(a) = 0, \quad (12)$$

for $d = 5, 7, 35$. The certificate checks 47 profile coordinates exactly, including rational ℓ^2 norms, and does not use floating-point comparison.

5.2 Interpretation

The fixture supplies the missing interface theorem at the finite algebraic level: the literal common-source map produces nontrivial nested-divisor Gram entries. It also prevents a stronger but unsupported conclusion. Because the cross terms are positive in the unit-weight fixture, coupling by itself does not imply a saving. In the physical problem the coefficients include $\mu(d) \log d/d$, the smooth function ψ , and the signed packet reassembly. Those ingredients may create cancellation, but that is a new estimate rather than a consequence of the lift identity.

6 Route evaluation and limitations

Route A is not applicable: this session concerns an analytic prime/twin-prime carrier, not a dynamical spectral family. On Route B, the structural threshold passes because the physical coupling map and its exact Gram are now explicit. The status ledger is

physical profile/emitter pullback	PROVED_EXACT
residue-lift gcd aliasing	PROVED_EXACT
frequency-intersection Gram	PROVED_EXACT_Finite
direct-sum physical replacement	REFUTED_SCOPED
literal V46 asymptotic Gram bound	OPEN
prime-shell reassembly	OPEN
arithmetic advance	NO
strict 1/400	UNPAID

Three limitations matter for the next step. First, the certificate uses $\psi = 1$ and omits the logarithmic scalar in order to keep the finite Gram integer-valued. Second, complete-period orthogonality is not an asymptotic estimate for an arbitrary interval; the exact finite-interval formula is Theorem 4.1. Third, the profile correction remains an affine term in (7); it has not been absorbed into a single source sequence.

The next natural problem is therefore to group the literal V46 kernel by shared rational frequencies, retain the signs and smooth weights, and prove a bound for the resulting joint clusters. If such clusters remain positive, the correct output is an obstruction rather than an advertised saving. Either outcome is more informative than applying a direct-sum Cauchy estimate before the coupling map is constructed.

7 Conclusion

TPC-213 resolves the operator-level ambiguity left by TPC-212. The physical residuals are generated by one common source, and the corresponding emitter pullbacks have an exact cross-divisor Gram. Chinese-remainder compatibility controls the residue lifts; rational-frequency intersections control the complete-period emitter Gram. The finite certificate exhibits the predicted nested-divisor cross terms and verifies the profile invariants exactly.

The result is a structural bridge, not a twin-prime theorem. It earns no arithmetic L^2 credit and leaves the strict 1/400 endpoint unpaid. The next theorem must estimate the joint pullback in the literal V46 range while preserving the smooth emitter, packet signs, zero-axis normalization, and prime-shell reassembly.

References

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